Week 11

1. Normal Distribution – dnorm()

```
x <- seq(-10, 10, by = 0.1)
y <- dnorm(x, mean = 2.0, sd = 1.5)
plot(x, y, main = "Density Normal Distribution",
xlab = "x", ylab = "Density", type = "l")
model <- lm(y ~ x)
abline(model, col = "red", lwd = 2)
```

2. Normal Distribution – pnorm()

```
x <- seq(-10, 10, by = 0.1)
y <- pnorm(x, mean = 2.0, sd = 0.5)
plot(x, y, main = "Cumulative Normal Distribution",
xlab = "x", ylab = "Cumulative Probability", type = "l")
model <- lm(y ~ x)
abline(model, col = "red", lwd = 2)
```

3. Normal Distribution – qnorm()

```
x <- seq(0, 1, by = 0.02)
y <- qnorm(x, mean = 2, sd = 1)
plot(x, y, main = "Quantile Normal Distribution",
xlab = "Probability", ylab = "Quantile", type = "l")
model <- lm(y ~ x)
abline(model, col = "red", lwd = 2)
```

4. Normal Distribution – rnorm()

```
y <- rnorm(50)
hist(y, main = "rnorm", xlab = "Values", col = "blue")
```

5. Binomial Distribution – dbinom()

```
x <- seq(0, 50, by = 1)
```

```
y <- dbinom(x, size = 50, prob = 0.5)
plot(x, y, main = "Binomial Distribution",
xlab = "Number of Successes", ylab = "Probability",
type = "h")
```

6. Binomial Distribution – pbinom()

```
x <- 0:51
y <- pbinom(x, size = 51, prob = 0.5)
plot(x, y, main = "Cumulative Binomial Distribution",
xlab = "Number of Successes", ylab = "Cumulative Probability",
type = "s")
```

7. Binomial Distribution – qbinom()

```
x <- qbinom(0.25, size = 51, prob = 0.5)
print(x)
plot(x, main = "Binomial Quantile")
```

week8

```
# Mean
x <- c(10, 20, 30, 40, 50)
result <- mean(x)
print(result)
# Median
x <- c(10, 20, 30, 40, 50)
result <- median(x)
print(result)
# Mode (user-defined function)
getmode <- function(a) {
```

```
uniq_a <- unique(a)
uniq_a[which.max(tabulate(match(a, uniq_a)))]
}
a <- c(10, 20, 10, 30, 20, 10, 40)
result <- getmode(a)
print(result)
# Covariance
x <- c(10, 20, 30, 40)
y <- c(50, 60, 70, 80)
result <- cov(x, y)
print(result)
boxplot(x, y, names = c("x", "y"), main = "Boxplot of x and y")
# Linear regression
x <- c(10, 20, 30, 40)
y <- c(50, 60, 70, 80)
result <- lm(y ~ x)
print(result)
plot(x, y, main = "Linear Regression", xlab = "x", ylab = "y", pch = 20)
abline(result, col = "red", lwd = 2)
```

week 9

2. Linear Regression

```
stdhrs <- c(1, 2, 3, 4, 5)
marks <- c(30, 40, 50, 60, 70)
student <- data.frame(stdhrs, marks)
print(student)
```

```
model <- lm(marks ~ stdhrs, data = student)
print(summary(model))
plot(stdhrs, marks,
main = "Linear Regression",
xlab = "Study Hours", ylab = "Marks",
pch = 20, col = "blue")
abline(model, col = "red", lwd = 2)
# Predict marks for the given study hours
new_student <- data.frame(stdhrs = c(6, 7))
print(predict(model, newdata = new_student))
```

3. Multiple Linear Regression

```
experience <- c(1, 2, 3, 4, 5)
qualification <- factor(c("SSC", "BTech", "MTech", "PhD", "MPhil"))
salary <- c(1, 2, 3, 4, 5)
employee <- data.frame(experience, qualification, salary)
print(employee)
model <- lm(salary ~ experience + qualification, data = employee)
print(summary(model))
plot(employee$experience, employee$salary,
main = "Experience vs Salary",
xlab = "Experience (years)", ylab = "Salary",
pch = 20, col = "purple")
abline(lm(salary ~ experience, data = employee), col = "red", lwd = 2)
```

4. Logistic Regression

```
students <- data.frame(
```

```
studyhours = c(1, 2, 3, 4, 5, 6, 7, 8),
pass = c(0, 0, 0, 1, 1, 1, 1, 1)
)
model <- glm(pass ~ studyhours, data = students, family = binomial())
print(summary(model))
students$probability <- predict(model, type = "response")
students$prediction <- ifelse(students$probability >= 0.5, 1, 0)
print(students)
plot(students$studyhours, students$probability,
main = "Logistic Regression",
xlab = "Study Hours", ylab = "Probability of Passing",
pch = 19)
lines(students$studyhours, students$probability, lwd = 2)
```

5. Poisson Regression

```
hrs <- c(1, 2, 3, 4, 5, 6, 7, 8)
ques <- c(2, 3, 4, 6, 8, 10, 13, 16)
data <- data.frame(hrs, ques)
print(data)
model <- glm(ques ~ hrs, family = poisson(), data = data)
print(summary(model))
predicted <- predict(model, type = "response")
print(predicted)
plot(data$hrs, data$ques,
pch = 19, xlab = "Study Hours",
ylab = "Questions Answered",
main = "Poisson Regression")
lines(data$hrs, predicted, lwd = 2)
```

week10

Time Series Analysis

```
sales <- c(100, 200, 300, 400, 500, 600, 700, 800, 900, 1000)
sales_ts <- ts(sales, frequency = 12)
print(sales_ts)
plot(sales_ts, type = "o",
main = "Time Series Analysis",

xlab = "Months", ylab = "Sales", lwd = 2)
```

7. Nonlinear Least-Squares Regression

```
year <- c(1, 2, 3, 4, 5, 6)
population <- c(100, 110, 125, 140, 160, 185)
model <- nls(population ~ a * exp(b * year),
start = list(a = 90, b = 0.1))
print(summary(model))
predicted <- predict(model)
print(predicted)
plot(year, population, pch = 19,
main = "Population Growth (Nonlinear Regression)",
xlab = "Year", ylab = "Population (thousands)")
lines(year, predicted, lwd = 2)
new_year <- data.frame(year = 8)
predicted_8 <- predict(model, newdata = new_year)
print(predicted_8)
```

8. Decision Tree Modeling (Regression Tree)

```
# Install these packages once if they are not already installed:
# install.packages("rpart")
# install.packages("rpart.plot")
library(rpart)
library(rpart.plot)
data(mtcars)
reg_tree <- rpart(mpg ~ hp + wt,
method = "anova",
data = mtcars)
rpart.plot(reg_tree, type = 2, digits = 3,
fallen.leaves = TRUE,
main = "Regression Decision Tree (Predicting MPG)")
```